6.5 Exponential Growth and Decay

Name _____ Period ____ Date _____

Algebra I 6.5 (30-minute plan) · Student sheet

I can…

- identify and write exponential growth and decay functions.
- find the percent rate of change of an exponential function.
- solve real-life problems, including compound interest.

Vocabulary · fill in the missing word

Exponential growth — $y = a(1 + r)^t$: a quantity _____ by the same percent each time period

Exponential decay — $y = a(1 - r)^t$: a quantity _____ by the same percent each time period

Growth / decay factor — _____

Compound interest — $y =$ _____

Part 1 · Notes and examples (fill in each note, then work the example)

Exponential growth: $y =$ _____ **Exponential decay:** $y =$ _____

Reading the model: a is the _____, and r is the _____ written as a decimal.

Example 1

For $y = 500(1.06)^t$, identify the initial amount and the rate of growth. Evaluate when $t = 3$.

Growth and decay factors: The growth factor is _____; the decay factor is _____.

Example 2

Does each function show growth or decay? Find the percent rate of change.

a. $y = 50(0.8)^t$ b. $y = 10(1.035)^t$

Compound interest: $y =$ _____

Example 3

A $9,000 deposit earns 6% annual interest compounded semiannually. Write a function for the balance and find it after 5 years.

Part 2 · Partner practice

Partner practice

A $2,000 laptop loses 20% of its value each year. Write a function and find its value after 3 years.